- Internet of Things (IoT)
  - Overview and Genesis
    - Traditional Internet vs IoT
      - Internet: Human-centric communication and information access
      - IoT: Interconnected physical objects collecting and exchanging data
      - First IoT-like Device: Coke machine at Carnegie Mellon University (1982)
      - Term Coined: Kevin Ashton at MIT Auto-ID Center (1999)
    - Core Characteristics
      - Dynamic and Self-Adapting: Automatic environment adjustment
      - Self-Configuring: Automatic network integration
      - Interoperable Protocol: Cross-manufacturer communication standards
      - Unique Identity: Distinct IP or MAC address for each device
  - Physical Design and Hardware
    - Tangible Components
      - Things: Smart thermostats and wearable fitness bands
      - Sensors: Temperature sensors in smart fridges
      - Actuators: Motors for automatic window blinds
      - Gateway Devices: Connecting local nodes to the cloud
    - Generic Block Diagram
      - Connectivity: USB Host and RJ45 Ethernet
      - Processor: CPU for sensor data and control
      - Memory: NAND/NOR Flash and DDR RAM
      - I/O Interfaces: UART, SPI, I2C, and CAN
  - Architectural Models
    - Three-Layer Architecture
      - Perception Layer: Physical sensors and actuators
      - Network Layer: Gateways and routers
      - Application Layer: User interfaces and software
    - IoT World Forum 7-Layer Model
      - Level 1: Physical Devices and Controllers
      - Level 2: Connectivity and Processing Units
      - Level 3: Edge Computing and Data Analysis
      - Level 4: Data Accumulation and Storage
      - Level 5: Data Abstraction and Access
      - Level 6: Application Reporting and Control
      - Level 7: Collaboration and Business Processes
    - Service-Oriented Architecture
      - Service Providers: Devices offering services
      - Service Consumers: Applications using services
      - Service Registry: Directory for dynamic discovery
      - Service Interface: RESTful APIs and MQTT topics
  - IoT Protocols Stack
    - Link Layer
      - IEEE 802.3: Wired Ethernet standard
      - IEEE 802.11: Wireless LAN (Wi-Fi)
      - IEEE 802.16: Metropolitan Area Network (WiMAX)
      - IEEE 802.15.4: Low-Rate Wireless Personal Area Network
    - Network Layer
      - IPv4: 32-bit addresses with 4.3 billion space
      - IPv6: 128-bit addresses with unlimited space
      - 6LoWPAN: IPv6 over low-power wireless networks
    - Transport Layer
      - TCP: Connection-oriented and reliable delivery
      - UDP: Connectionless and fast transmission
    - Application Layer Protocols
      - HTTP: Client-server request-response protocol
      - CoAP: Constrained application protocol over UDP
      - MQTT: Lightweight publish-subscribe messaging
      - XMPP: XML-based communication protocol
  - Logical Design and Communication
    - Communication Blocks
      - D2D: Direct device-to-device communication
      - D2G: Device-to-gateway communication
      - D2C: Direct device-to-cloud connection
      - G2C: Gateway-to-cloud aggregated data transfer
    - Communication Models
      - Request and Response: Stateless client-server model
      - Publisher-Subscriber: Broker-managed topic distribution
      - Push-Pull: Event-driven push vs request-driven pull
      - Exclusive Pair: Dedicated persistent connection
    - Communication APIs
      - REST-Based API: Resource access using HTTP methods
      - WebSocket API: Bi-directional full-duplex connection
  - Applications and Impact
    - Domain Applications
      - Smart Home: Automated thermostats and smart lighting
      - Industrial IoT: Smart factory predictive maintenance
      - Smart Agriculture: Soil moisture sensors and irrigation
      - Smart Cities: Automated waste management
    - Impact Areas
      - Manufacturing: Improved productivity and reduced downtime
      - Healthcare: Remote Patient Monitoring (RPM)
      - Economy: Cost reduction and revenue growth
  - Challenges and IT Convergence
    - Core Challenges
      - Security and Privacy: Vulnerability to hacking and breaches
      - Interoperability: Fragmented standards across vendors
      - Scalability: Managing billions of devices
      - Power Consumption: Energy efficiency for battery devices
    - IT and IoT Convergence
      - Data Flow: Real-time IoT data fed into IT systems
      - Infrastructure: Hybrid cloud, edge, and device integration
      - Benefits: Real-time decisions and operational efficiency